

Sara Aminpour

PH.D. STUDENT IN ELECTRICAL AND COMPUTER ENGINEERING

(405) 408-5688 | sara.aminpour@ou.edu | Norman, OK 73071

Ph.D. student and graduate research/teaching assistant at the University of Oklahoma, working at the intersection of quantum computing and artificial intelligence in the INQUIRE Laboratory. My work involves reviewing recent literature, designing and running experiments, and contributing to the development of quantum-enhanced algorithms. I collaborate closely with faculty and graduate students, assist in writing and editing research papers, and help ensure experiments are well-documented and reproducible.

Skills

- Python programming
- Quantum computing (Qiskit)
- Scientific writing & L^AT_EX
- Literature reviews
- Research and analysis
- Teaching assistance
- Analytical thinking
- Teamwork and collaboration
- Multitasking and organization
- Microsoft Office

Experience

INSTRUCTOR — CYBERSECURITY ESSENTIAL TRAINING WORKSHOP

Norman, OK

University of Oklahoma, Gallogly College of Engineering

Summers 2025 & 2026

- Served for two consecutive years (2025–2026) as an instructor for the OU *Cybersecurity Essential Training* workshop, sponsored by the **Oklahoma Office of Homeland Security** and **FEMA**.
- Delivered hands-on sessions on Quantum in Cybersecurity, introducing participants to post-quantum cryptography, quantum key distribution (QKD), and quantum attack models.
- Developed and presented technical content bridging classical and quantum cybersecurity concepts, tailored to professional audiences with varying levels of prior knowledge.
- Collaborated with INQUIRE Lab team members to design interactive sessions, real-world case studies, and demonstrations using quantum simulators and cryptographic tools.

TEACHING ASSISTANT

Norman, OK

University of Oklahoma

August 2024 – Present

- Supported instruction in key topics such as PLC, CLICK programming software, and LabVIEW programming, assisting over 20 students across multiple semesters.
- Led lab sessions, graded assignments and lab reports, and provided one-on-one guidance to reinforce concepts in automated measurement systems.
- Collaborated with course instructors to update lab materials, improve experiment design, and implement tools for more effective student assessment and feedback.

RESEARCH ASSISTANT

Norman, OK

University of Oklahoma

January 2024 – August 2024

- Conducted literature reviews and data-driven analyses on advanced topics in quantum computing and AI algorithms, contributing to ongoing research publications and grant proposals.
- Designed and implemented simulation experiments using Python, Qiskit, and hardware modeling tools, supporting the evaluation of quantum-accelerated cryptographic protocols.
- Collaborated with a multidisciplinary team to draft technical reports, visualize results, and present findings in research group meetings and academic seminars.

Education

DOCTOR OF PHILOSOPHY (PH.D.) IN ELECTRICAL AND COMPUTER ENGINEERING

Norman, OK

University of Oklahoma

In progress

Relevant Coursework: Quantum Mechanics II; IC Fabrication I; IC Fabrication II

Awards

- GSS Research Award — Fall 2026
- ECE Travel Award — Spring 2026
- GCoE Travel Award — Fall 2025
- GSS Research Award — Fall 2025

Service & Leadership

ECE Senator, Graduate Student Senate

2025 – 2026

University of Oklahoma — Academic Affairs Committee

- Represented the School of Electrical and Computer Engineering in the OU Graduate Student Senate, serving on the Academic Affairs committee.

Publications

Journal Articles

1. **Aminpour, S.**, Banad, Y. M., and Sharif, S. S. “Strategic Data Re-Uploads: A Pathway to Improved Quantum Classification—Data Re-Uploading Strategies for Improved Quantum Classifier Performance.” *Entropy*, 28(5), 550, 2026.
2. **Aminpour, S.**, Sharif, S., and Banad, M. “Development of Hybrid Quantum Classifiers for Realistic Classification Tasks.” *Journal of Physics Communications*, 10(2), 025004, 2026.

Conference Proceedings & Abstracts

3. **Aminpour, S.**, Banad, Y., and Sharif, S. “Learning with a Single Qubit: Classical Training of Quantum-Inspired Classifiers for Structured Tasks.” *Quantum Computing, Communication, and Simulation VI*, Proc. SPIE 13919, pp. 131–135, 2026.
4. **Aminpour, S.**, Banad, Y., and Sharif, S. “Exploring Quantum Advantage in Classification: Single-Qubit vs. Entangled Systems.” *Quantum Computing, Communication, and Simulation V*, Proc. SPIE 13391, pp. 170–173, 2025.
5. **Aminpour, S.**, Banad, Y., and Sharif, S. “Quantum Classifier with Iterative Re-Uploading for Universal Classification: Performance Evaluation and Insights.” *APS March Meeting Abstracts*, FF00.034, 2024.
6. **Aminpour, S.**, Banad, Y., and Sharif, S. “Quantum Machine Learning Performance Analysis: Accuracy and Efficiency Trade-offs in Linear Classification.” *Frontiers in Optics* (Optica Publishing Group), JW5A.72, 2024.
7. Sharifi, S., and **Aminpour, S.** “Re-Uploading Classical Data Points Using Quantum MP Neural Network.” *APS March Meeting Abstracts*, G00.289, 2023.

Preprints

8. **Aminpour, S.**, Banad, M., and Sharif, S. “Boosting Quantum Classifier Efficiency through Data Re-Uploading and Dual Cost Functions.” Research Square preprint, 2025.